AE 158 – Introduction to Information Technology (2,1,2)

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Course Outline

- INTRODUCTION TO COMPUTERS
- COMPONENTS OF AN INFORMATION SYSTEM
  - Computer software
  - Computer hardware
- BASICS OF NETWORK CONNECTIVITY AND INTERNET USAGE
- INTRODUCTION TO COMPUTER PROGRAMMING LANGUAGES

LECTURE ONE

- Introduction and History of Computers

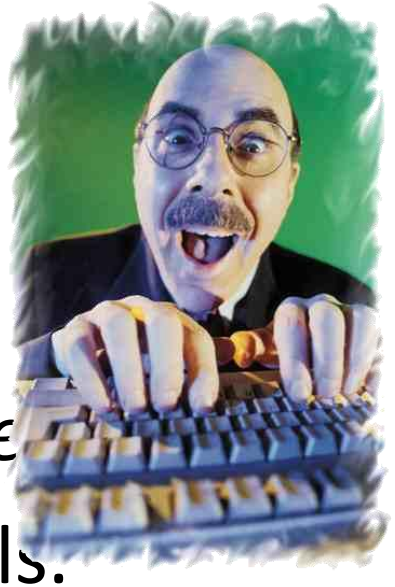
- Computer systems, its components and types of computers

- Characteristic of computers and application of computers

Introduction and History of Computers

• Why do I need IT knowledge?

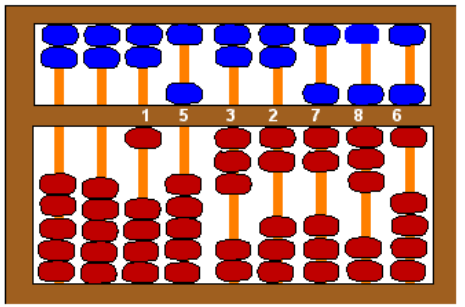
- 83% think computer competency skills are important or very important in the hiring decision.
- 96% want basic *word processing* skills.
- 93% want *e-mail* experience.
- 86% want basic *spreadsheet* skills.
- 83% want basic *database* skills.
- 75% want basic *presentation software*
- 64% want *internet* and *searching* skills.



From: T.H.E. Journal - September 1997: What Computer Skills Do Employers Expect From Recent College Graduates?

Historical Development of Computers

- ABACUS -1st computer to be invented in China about 600 BC.



- **John Napier Bones** (1550 – 1617)
  - invention of logarithm (by 1672, the principle had been applied to a calculation device called the SLIDE RULE.
- **Blaise Pascal (1642)**
  - first adding machine (precursor of the digital computer)
  - Gottfried Wilhelm von Leibniz (1970s) improved on this machine by devising one that could also multiply.

Historical Development of Computers

- **Joseph Marie Jacquard**
  - Inventor of the automatic loom, used for the weaving of complicated designs.
- **R. Herman Holerith (1880s) –**
  - Inventor of modern punched cards. International Business Machine (IBM) *came out through this man*.
- **Charles Babbage** (19th century)-
  - worked on the principle of modern digital computers.
  - He conceived a number of machines, such as the Difference Engine, that were designed to handle complicated mathematical problems.

Computer systems, its components and types of computers

Introduction

Information Technology (IT)

- the study or use of systems (especially computers and telecommunications) for storing, retrieving, and sending information.
- A broad term which covers all aspects of the use of computer technology.
  - It integrates *computing technology and information processing.*

Computer

Compute	ter *(Latin)*
Calculating	Machine

Computer (to Compute)

- Normally Considered to be calculating device

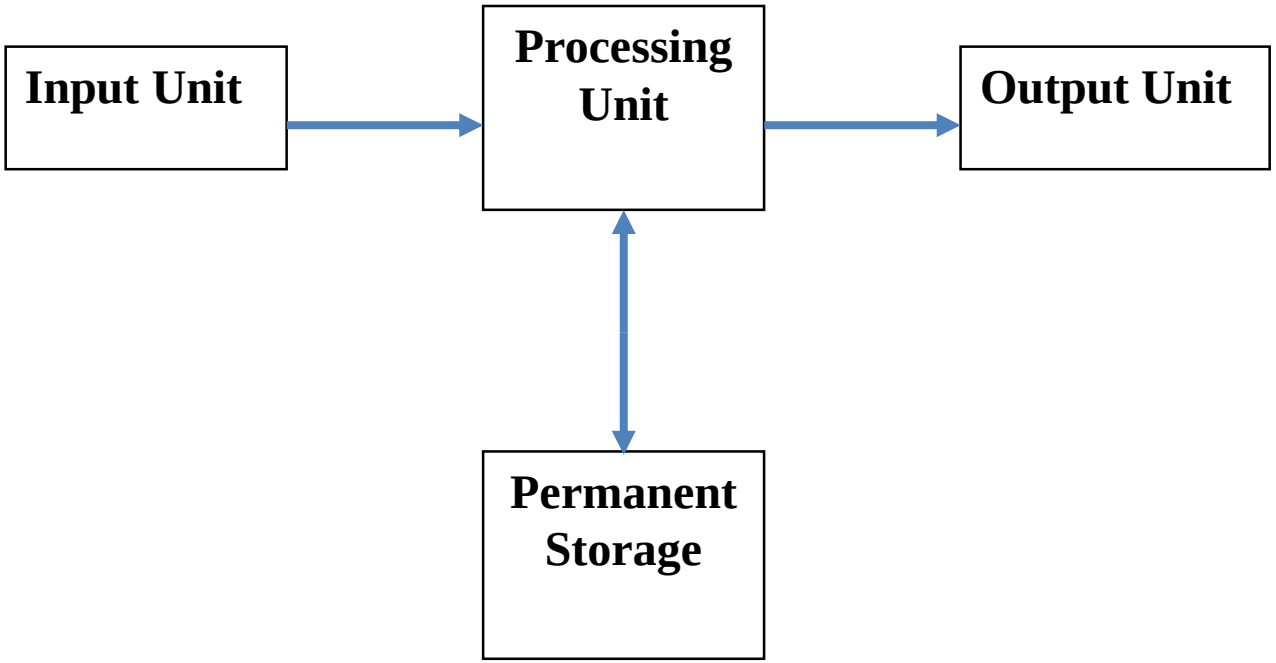
- **But today it performs more then 80% non numeric nature works**

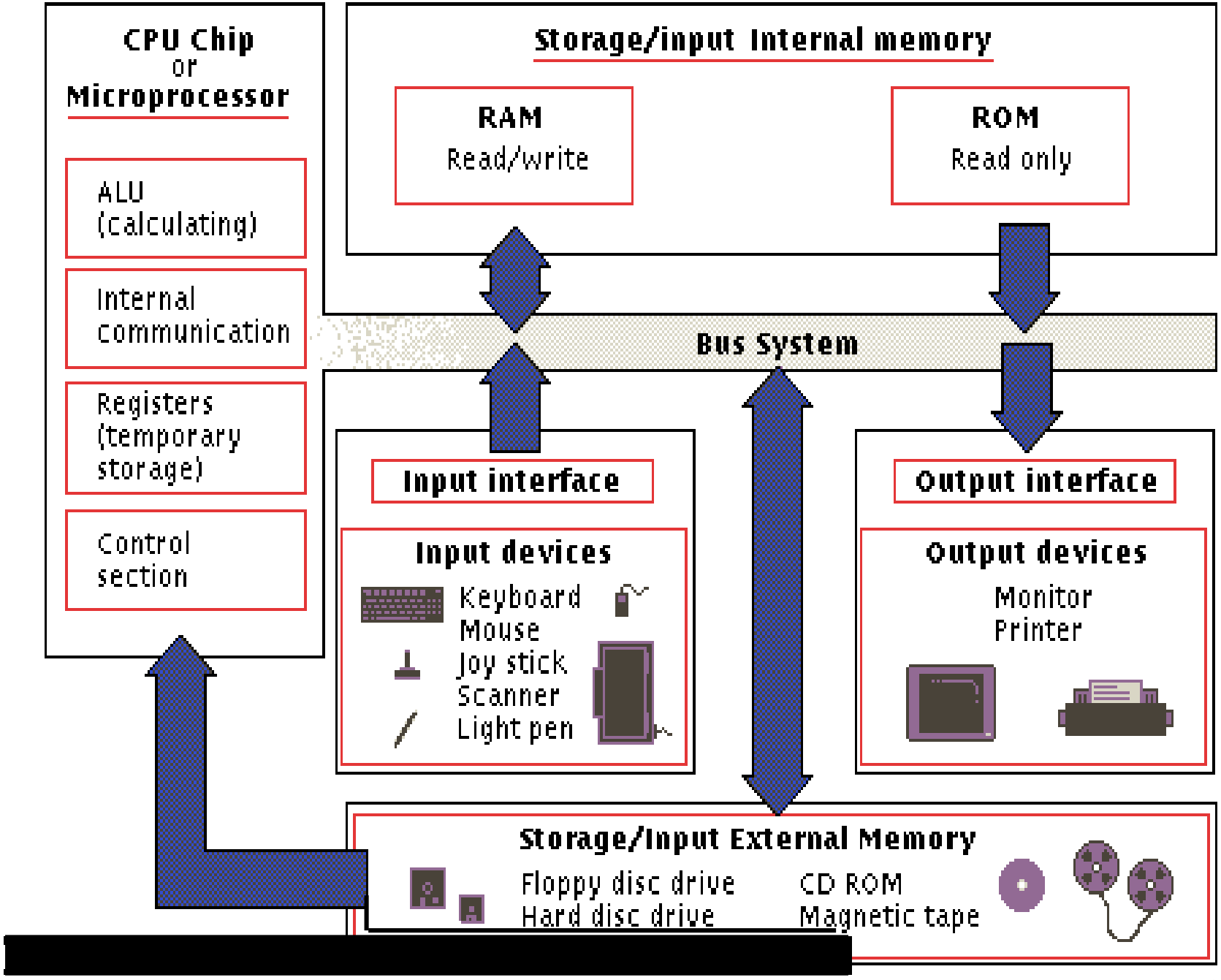


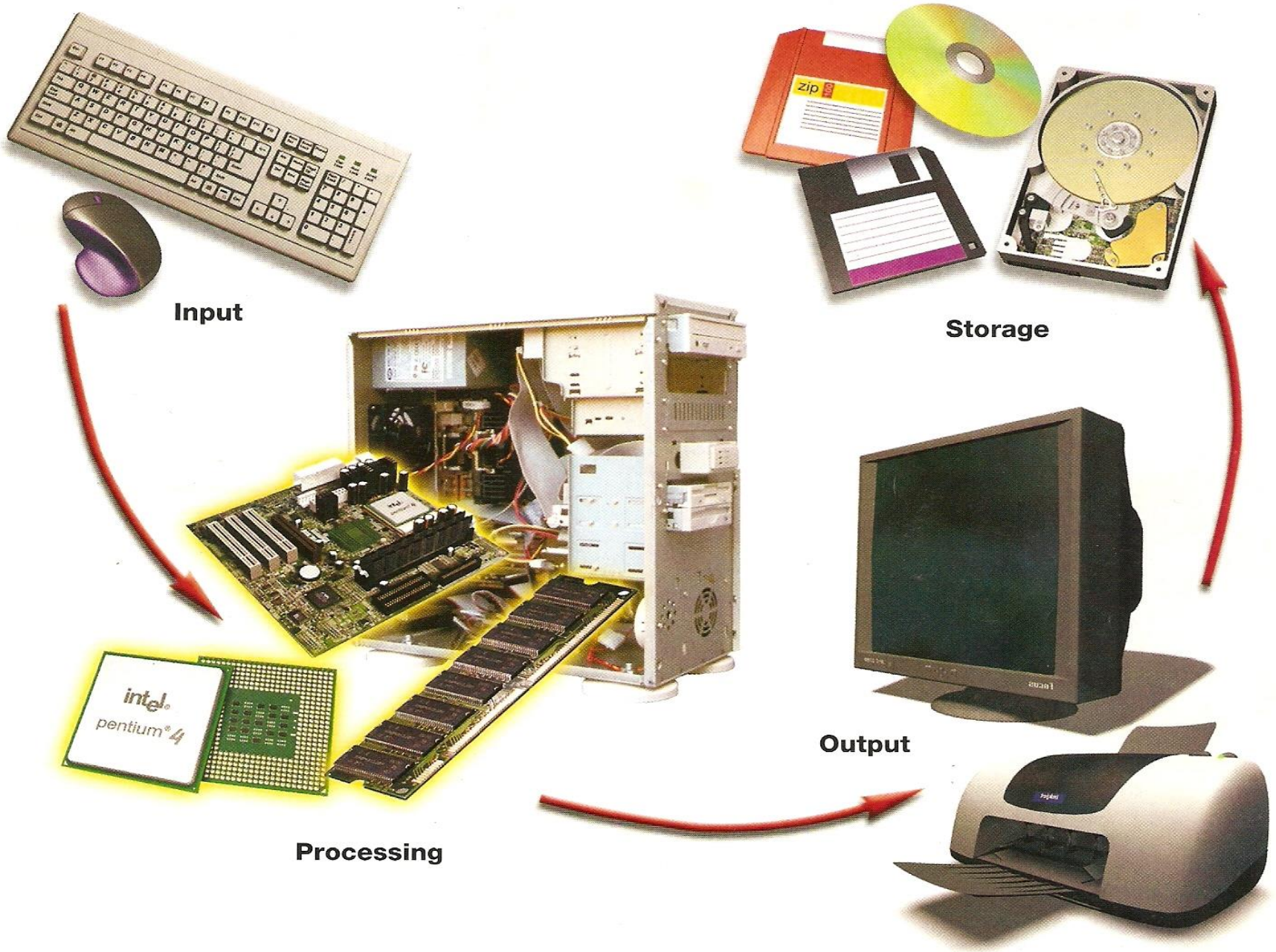
Hence a Computer can be defined as:

- Electronic device that works under the control of stored programs, automatically accepting, storing, and processing data to produce information.
- A computer therefore functions by
  - accepting input data
  - processing the data
  - Storage of the data internally, before, during and after processing
  - Output the data for external use
- All is achieved through communication network called a *bus* links all the units together

Computer System and its components

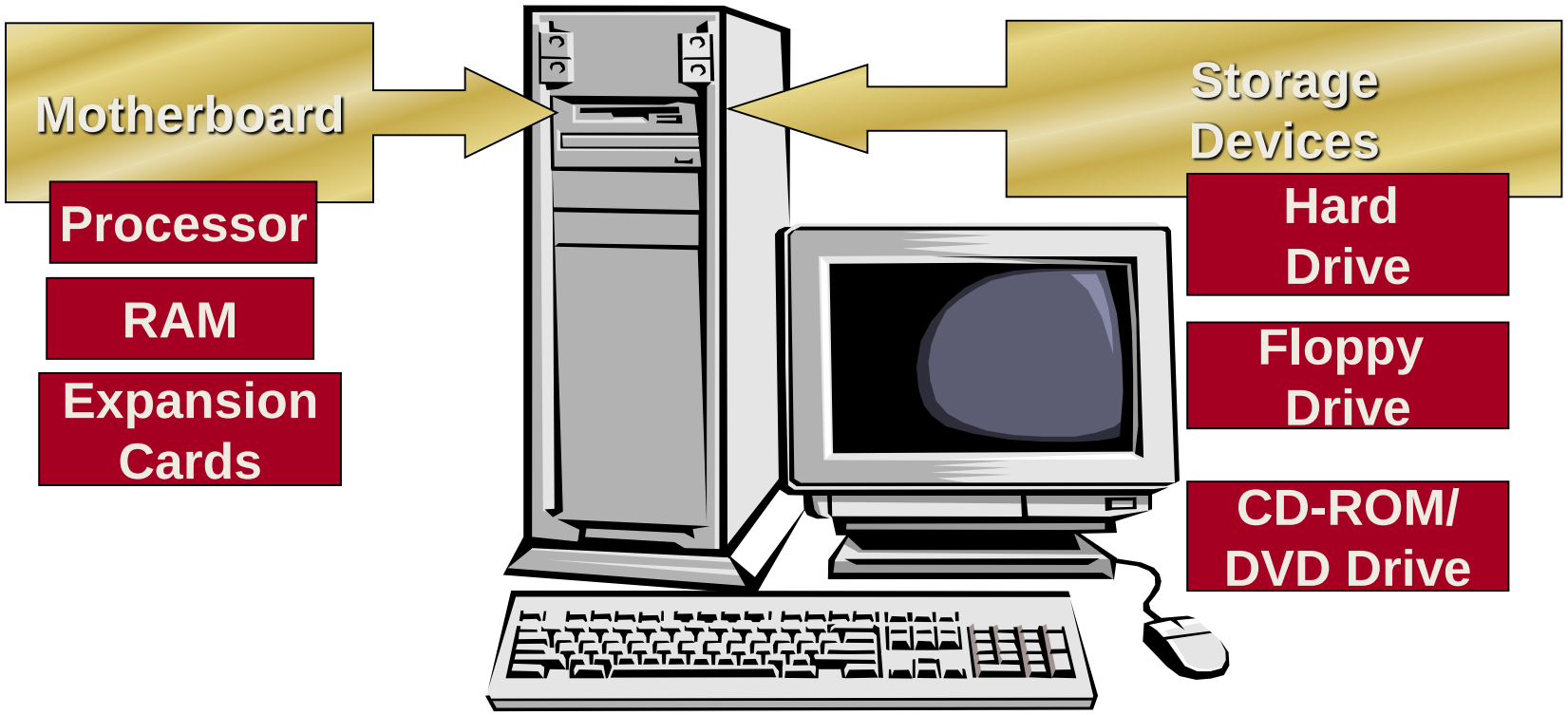



CPU Chip
or
Microprocessor
ALU
(calculating)
Internal
communication
Registers
(temporary
storage)
Control
section
Storage/input Internal memory
RAM
Read/write
ROM
Read only
Bus System
Input interface
Input devices
Keyboard
Mouse
Joy stick
Scanner
Light pen
Output interface
Output devices
Monitor
Printer
Storage/Input External Memory
Floppy disc drive
Hard disc drive
CD ROM
Magnetic tape


Input
Storage
Processing
Output
intel
pentium 4

Components of a computer

- *Know the main parts of a personal computer such as: central processing unit (CPU), common input and output devices, types of memory. Understand the term peripheral device.*



Central Processing Unit

- brain of the computer also called **processor**
- the part of the computer that runs the program
  - receives all program instructions
  - performs the arithmetic and logical operations
  - controls all the other computer components
- made up of millions of electronic components called
  - transistors, capacitors and resistors.

Input devices

- Input devices are components which are used to
  - feed commands (text, images, and sound) and data into the computer.

- Examples include devices such as keyboard, mouse, microphone, modem, and drawing tablet.

Output devices

- Output devices are devices that the computer uses to show us the results of the processed data

- These include the VDU (monitor), printers and speakers

Peripheral devices

- A computer system comprises the case which houses the motherboard, CPU, hard drives, power supply and various other components.

- Devices such as printers, monitor, keyboard and mouse which are connected to the main unit by various types of cables and connectors are called **peripheral devices**.

Memory

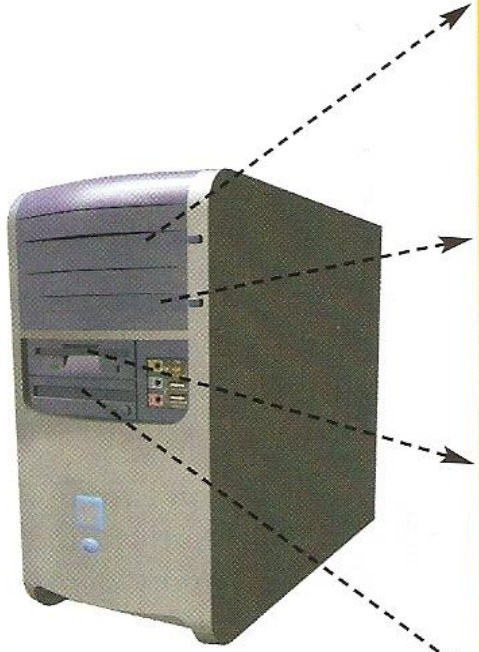
- Two general categories of memory:
  - primary and secondary
- **Secondary memory**
  - devices that are able to store data and programs even when the power is off.
  - includes devices such as hard drives, floppy drives, tape drives, CD drives and DVD drives.
- **Primary memory**
  - associated with the actual working of the computer. This includes memory that holds the start-up routines as well as the current program and data it is working with.
  - various forms of primary memory: RAM, ROM and Cache memory.

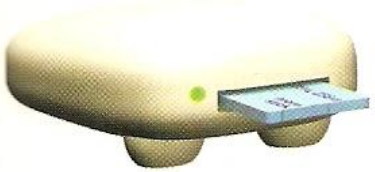
- **RAM or Random Access Memory**
  - holds the current running program and its associated data.

- **ROM or Read Only Memory**
  - contains certain key routines (small programs).
  - One example, is the set of start-up routines. These take control of the computer when you switch on and ensure that the computer boots-up.
  - Booting-up is the process of starting the computer up so that it is able to load and run computer programs.

- **Cache memory**
  - very high speed memory that is used by the CPU in executing the individual instructions of the program.
  - It is used to hold items such as instructions that are next in line to be executed and data that is likely to be needed by the CPU.

➢**Storage devices:** Storing of processed data





	STORAGE MEDIUM	CAPABILITIES	STORAGE CAPACITY
	DVD DVD+RW	Read-only Read and Write	9.4 GB
	CD CD-RW	Read-only Read and Write	700 MB
	Floppy Disk	Read and Write	1.44 MB
	Zip Disk	Read and Write	100 MB, 250 MB, or 750 MB
	Flash Memory Cards	Read and Write	16 MB to 1 GB

Types of Computers

- **Different categories of computers**
- **Classification based on**
  - operational principle
    - Analog, Digital and Hybrid computers
  - physical structure and the purpose of use and
  - capacity, speed and reliability
    - Mainframe Computer
    - Supercomputers
    - Microcomputer
    - Mini computer
    - Tablet computers

Types of Computers

- Analog computers
  - electronic or hydraulic device that is designed to handle input in terms of, for example, voltage levels or hydraulic pressures, rather than numerical data.
  - carry out arithmetic and logical operations by manipulating and processing data which you input such as the weight, temperature, voltage, power, density.
  - In a typical electronic analog computer, the inputs are converted into voltages that may be added or multiplied using specially designed circuit elements.
  - The simplest analog calculating device is the slide rule

Types of Computers

- Digital computers
  - Everything that a digital computer does is based on one operation.
    - the ability to determine if a switch, or “gate,” is open or closed.
  - in any of its microscopic circuits the computer can recognise only two states :
    - on or off,
    - high voltage or low voltage,
    - or-in the case of numbers-0 or 1.
  - solve mathematical problems at a tremendous speed but cannot perform bulk operations simultaneously

Only 2 states possible



On Off

0 1 0 0 0 0 0 1 ← A

Off On Off Off Off Off Off On

Types of Computers

- **Hybrid**
  - combination of elements of both analog and digital computers
  - usually used for problems in which large numbers of complex equations, known as time integrals are to be computed
  - data in analog form can also be fed into a digital computer by means of an analog- to-digital converter, and the same is true of the reverse situation

Types of Computers

Mainframes

- largest and most powerful of computers that handles processing for many users simultaneously
- Users connect to the mainframe using terminals and submit their tasks for processing by the mainframe.
  - A terminal is a device that has a screen and keyboard for input and output, but it does not do its own processing. (also called dumb terminals since they can't process data on their own)
- used in situations where a company wants the processing power and information storage in a centralized location



Types of Computers

Supercomputer

- mainframe computer that has been optimized for speed and processing power.
- have the largest capacity in terms of data storage and processing speed
- found in large corporate institutions, research organisations, government ministries and tertiary academic institutions.
- used for extremely calculation-intensive tasks such simulating nuclear bomb detonations, aerodynamic flows, and global weather patterns



Types of Computers

Microcomputer

- most frequently used computer also known by the name “Personal computers” (PC)

- PCs come in three sizes:
  - Desktop and tower units
  - Laptops
    - Notebook computers
    - Sub-notebook computers



  - Pocket PCs or Handheld computers
    - Electronic organisers
    - Palm top computers
    - Pen computers/Personal digital assistants





Types of Computers

Mini computer

- medium-sized computers also known as midrange computers
- used by medium-sized organisations and departments within larger organisations for the maintenance of accounts and finances
- much less expensive than the mainframes

Characteristic of computers and application of computers

Characteristics of a Computer

- **Speed**

  The fastest computer in 1993, the CM-5, could perform 131 billion operations per second.

  With a pocket calculator this number of operations would take 41 000 years.

- **Memory**

  Human forget important information with time

  A Computer can store and recall every piece of information.

  A computer forget or looses information only when it is asked.

Characteristics of a Computer

- **Diligence**

  A Computer is free from monotony, tiredness, lack of concentration etc.

- **Versatility**

  Adapting easily to different subjects or occupations; skilled in many subjects

- **Accuracy**

  Errors can occur in a computer, but these are mainly due to human rather than technological weakness, that is, due to imprecise thinking by the programmer or due to incorrect data

The Use of Computers in Everyday Life

- Computers are ideal for repetitive work requiring speed and accuracy
- Some examples include:
  - Corporate data processing including functions such as sorting, selecting, summarising
  - Analysis of census and other demographic data
  - Administration of the national revenue system
  - Statistical analysis
  - Corporate accounting functions
  - Creation of animations for films
  - Weather forecasting
  - Forensic analysis such as DNA and fingerprint matching
  - Manufacture of electronic components and circuitry

Corporate Application

- *Some of the uses of large-scale computer applications in business includes:*
  - *business administration systems,*
  - *airline booking systems,*
  - *insurance claims processing and*
  - *online banking*

- All organisations, no matter their size, can benefit from computer applications.
- Examples found in business include:
  - Office application suites such as OpenOffice.org, Koffice, StarOffice or Microsoft Office for creating documents, spreadsheets and presentations.
  - Inventory systems for keeping track of stock.
  - Desktop publishing packages such as Microsoft Publisher and Page Maker for creating newsletters and press releases.
  - Airline booking systems which manage large amounts of data and reservation details and also have the flexibility to handle frequent changes to bookings.
  - Insurance claims systems to manage the processing and payment of claims.
  - Online banking systems enable corporates and individuals to have easy access to funds transfer and account maintenance.

Public Sector Applications

- *Some of the uses of large-scale computer applications in government includes:*
  - *public records systems (census, vehicle registration),*
  - *revenue collection,*
  - *electronic voting*
  - *National Identity System*

Health Sector Applications

- *Some of the uses of large-scale computer applications in hospitals/healthcare includes:*
  - *patient records systems,*
  - *ambulance control systems,*
  - *diagnostic tools and instruments,*
  - *specialist surgical equipment.*

Education Sector Applications

- *Some of the uses of computer applications in education includes:*
  - *student registration and timetabling systems,*
  - *computer-based training (CBT),*
  - *automated examinations*
  - *distance learning,*
  - *Homework using the internet.*

Teleworking Applications

- Teleworking means literally doing work at a distance.
  - This means instead of going into an office, you work from home, a holiday cottage, another country or any other location.
- When you do teleworking, you can be thought of as having a **virtual office**
- Teleworking makes use of email, the Internet and Virtual Private Networks (VPNs)
- Occupations uses teleworking include:
  - journalists, writers, computer programmers, graphic artists, consultants and representatives.

Advantages of Teleworking

- **Professionals**
  - do not waste time on commuting between home and work.
  - are free to undertake work where ever they are.
  - greater ability to focus on one task.
  - have flexibility to arrange their work time according to their needs and inclinations.
  - generally have tax advantages as they can claim business expenses.
- **Organisations**
  - save on office space, equipment and facilities.

Disadvantages of Teleworking

- **Professionals**
  - may suffer from lack of human contact.
  - need to be highly self-disciplined.

- **Organisations**
  - will not get the benefits of teamwork.
  - will have less control over workers.

Other Applications

- **Buildings**
  - Architects use computer animated graphics to experiment with possible exteriors and to give clients a visual walk-through of their proposed buildings.

- **Energy**
  - Energy companies use computers to locate oil, coal, natural gas and uranium. With the use of these technological machines, these companies can figure out the site of a natural resource, its concentration and other related figures.

Applications of Computers

- **Retailing**
- Products from meats to magazines are packed with zebra-striped bar codes that can be read by the computer scanners at supermarket checkout stands to determine prices and help manage inventory.

- Thus, a detailed receipt of the groceries can be made, which is useful for both the customer and the retail store, especially for the stock control system.

- This is referred as POS (Point of Sale) transaction where a precise account of all the stocks available is recorded and manipulated.

Applications of Computers

- **Law Enforcement**

Recent innovation in computerized law enforcement include national fingerprint files, a national file on the mode of operation of serial killers, and computer modeling of DNA, which can be used to match traces from an alleged criminal's body, such as blood at a crime scene. In addition, computers also contain a complete databases of all the names, pictures and information of such people who choose to break the law.

Applications of Computers

- **Transportation**
  - Computers are used in cars to monitor fluid levels, temperatures and electrical systems.
  - An important part is the air control traffic systems, where computers are used to control the flow of traffic between airplanes which needs a lot of precision and accuracy to be dealt with.

Applications of Computers

- **Money**
  - –Computers speed up record keeping and allow banks to offer same-day services and even do-it yourself banking over the phone and internet.
  - –Computers have helped fuel the cashless economy, enabling the widespread use of credit cards, debit cards and instantaneous credit checks by banks and retailers.
  - –There is also a level of greater security when computers are involved in money transactions as there is a better chance of detecting forged cheques and using credit/debit cards illegally etc.

Applications of Computers

- **Manufacturing Industries**

•Computers have made their way towards jobs that were unpleasant or too dangerous for humans to do, such as working hundreds of feet below the earth or opening a package that might contain an explosive device. In other industries, computers are used to control the production of resources very precisely.

•All robots and machinery are now controlled by various computers, making the production process faster and cheaper. All the stages of manufacturing, from designing to production, can be done with the use of computer technology with greater diversity.

Applications of Computers

- **Scientific Research**
  - This is very important for mankind and with the development of computers; scientific research has propelled towards the better a great deal.
  - Because of high-speed characteristics of computer systems, systems, researchers can simulate environments, emulate physical characteristics and allow scientists to proof of their theories in a cost-effective manner.
  - Also many test lab animals are spared since computers have taken over their roles in extensive research.

Applications of Computers

- There are so many applications of computers, that it is impractical to mention all of them. Computers are now becoming faster, more reliable, effective and whole lot cheaper than they had been ever before.